

High-Availability Reference Architecture

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MessageFoundry runs reliably as a single process, and scales to a highly available pair when you need to survive a host failure. This reference describes the supported HA model — **active/passive** (one leader, one or more warm standbys) — how leadership is coordinated, what the operator supplies, and what failover looks like.

The defining property of this design is that **all coordination lives in the shared SQL store**. There is no separate clustering broker, no Windows Server Failover Clustering (WSFC), and no Active Directory dependency. If you can stand up a server-class database and a basic load balancer, you can run MessageFoundry highly available.

Single-node vs. two-node active/passive

	Single node	Two-node active/passive
Processes	One engine	Two (or more) identical engines
Store	SQLite, PostgreSQL, or SQL Server	PostgreSQL or SQL Server (shared)
Who runs the message graph	The one node	The leader only
Failure of a host	Interface is down until restarted	A standby is promoted and resumes
Operator load balancer	Optional	Required (floating VIP / L4 LB)
Coordination	None needed	App-level leader lease in the store

Single-node operation is the default and is the same whether or not clustering exists in the build — clustering is strictly opt-in. Turn it on by enabling the cluster on a server-class store; every node then coordinates so that **exactly one** node — the leader — runs the message graph, and a standby takes over on failure.

Two-node active/passive is the supported HA topology. (A horizontal active/active scale-out model is **not** offered.)

Requirements

A clustered deployment requires, and enforces at configuration load:

- A **server-class shared store** — PostgreSQL or SQL Server. SQLite is a single-file, single-node store and cannot back a cluster; the cluster needs the shared membership table and row leases that a server database provides. Both PostgreSQL and SQL Server run the same active/passive leadership lease and are at parity.
- **Every node points at the same database** (same server, database, and schema) and runs the **same configuration directory**.

- A **connection-pool size with headroom** — a clustered node drives concurrent background work (membership and lease renewal, the leader reclaim sweep, and the per-stage workers), so give the pool room above its working connections.

The same `serve` command starts on each host. Nodes self-identify; you only need to pin a node identity for a stable identity across restarts.

The leader lease (no WSFC, no AD)

Leadership is an **application-level lease stored as a row in the shared database** — not an OS-level cluster service. This is what removes the WSFC/AD requirement: the database is the single source of truth for who leads.

How it works at a high level:

- **Exactly one node holds the leader lease** and is the leader. The leader renews the lease on a short heartbeat interval; the expiry is computed on the **database's own clock**, so node-to-node clock skew never affects who may lead.
- **A standby acquires leadership only after the lease has expired.** A standby never preempts a healthy leader.
- **A leader that cannot renew self-fences.** If the current leader cannot refresh its lease within its fence timeout — shorter than the lease TTL — it stops acting as leader *before* the lease can expire and a standby can acquire it. This is the split-brain guard: a network-partitioned or stalled old leader provably stops processing before any standby is promoted.
- **On a clean stop, the leader releases its lease immediately**, so a standby takes over promptly rather than waiting out a timeout.

Three timing values govern this, and the configuration enforces the invariant `heartbeat < fence timeout < lease TTL`. The defaults trade a roughly half-minute crash-failover window for ample tolerance of a momentarily slow database or garbage-collection pause; you can scale all three down proportionally for faster failover at the cost of less slack.

What the leader does (and the standby doesn't)

The wired message graph — **all** inbound listeners (MLLP, TCP, File, and the rest) **and** the router, transform, and delivery workers — runs **only on the leader**.

A **warm standby binds no listener ports and runs no workers**. It stays warm by maintaining its membership heartbeat and converging shared state (reference data, configuration version, transform state), and it brings the full graph up the moment it acquires leadership — and tears it down if it loses leadership. The graph supervisor polls leadership on a short interval, so a demotion or fence promptly stops a node from accepting new inbound work.

Singleton background work — retention purges, the lease-reclaim sweep, and polling of any **shared** external source (a watched directory, a database-poll table, a remote directory) — is **leader-gated** so it never double-executes.

Because only one node ever processes a given lane, **per-lane FIFO ordering is naturally preserved**, and it survives failover: a promoted leader reclaims a crashed predecessor's stranded in-flight work before delivering anything newer on that lane, so a later message can never jump ahead of an interrupted earlier one.

The floating VIP / L4 load balancer (operator-supplied)

Clients reach "the engine" through a **floating VIP or Layer-4 load balancer that you supply**, not a fixed hostname — so failover is transparent to senders apart from a reconnect. MessageFoundry does not ship this component; it is part of your network.

The mechanism is deliberately simple and relies on the leader-gating above:

- **For each inbound port, configure a TCP health check** — a plain TCP connect to that listener port.
- **Only the leader binds the port**, so the health check passes **only on the leader**. The VIP therefore routes inbound traffic to the leader automatically.
- **On failover, the new leader binds the port** and the old leader's socket closes, so the VIP follows the leader with no manual intervention.

For MLLP/TCP senders, configure partners to **reconnect on drop** (standard MLLP client behavior): on failover they see a connection drop and reconnect through the VIP to the new leader.

The engine API (the console / IDE control and read plane) is available on **every** node, so an API-facing VIP can health-check the unauthenticated liveness endpoint. To pin operations to the leader, the cluster exposes read-only status endpoints (this node's role — leader, standby, or single-node — and the cluster-wide membership with the live leader and lease holder); the monitoring console surfaces the current leader from these.

DB-tier HA is delegated to the database

MessageFoundry does **not** replicate the store itself. **High availability of the database tier is delegated to the database**, using its native mechanism:

- **PostgreSQL** — streaming replication with your chosen failover tooling.
- **SQL Server** — Always On availability groups.

Point all engine nodes at the database's HA endpoint (the replication/failover address), and the database's own failover handles store-tier resilience independently of engine-tier leadership.

Failover behavior

Failover is **not instantaneous** — plan for a promotion window rather than zero downtime:

- **Clean stop (planned switchover or graceful shutdown)**. The leaving leader releases its lease, so a standby is promoted on its next heartbeat — failover is prompt.
- **Crash or network partition**. The dead leader's lease ages out, and a standby acquires once it expires (up to the lease TTL). A partitioned old leader self-fences within its fence timeout — shorter than the TTL — so it stops processing before the standby is promoted.

Across the window, no message is lost:

- **In-flight rows are protected by per-row leases** — a standby reclaims only rows whose lease has expired.
- **On promotion, the new leader immediately recovers the predecessor's stranded in-flight work**, so delivery resumes at once rather than waiting out per-row lease timeouts.
- **Ordering is preserved** — the stranded head of each lane is reclaimed before newer work on that lane.
- Delivery is **at-least-once with idempotent re-runs**: a message interrupted mid-delivery is re-delivered once its lease frees, so **downstream endpoints should be idempotent**.

Operational guidance

- **Keep node clocks synced (NTP)**. Leadership is decided on the database clock, but per-row leases use node wall-clock, so reasonably synchronized clocks keep lease expiry consistent across nodes.
- **Deploy identical configuration to every node**. Each node loads its graph from its own configuration directory; the cluster coordinates the reload *version*, not the files. Keep the directories identical.
- **Apply configuration changes as a coordinated restart, not a rolling one**, so nodes never run divergent graphs across the change window.

Transport security still applies

HA does not change the transport posture. MessageFoundry binds loopback by default and **fails closed** on a non-loopback API or MLLP bind without TLS. When you expose a listener off loopback — which any real data-plane feed requires — enable TLS on that channel (the API supports in-process TLS or a trusted upstream terminator; MLLP supports per-connection TLS and optional mutual TLS). Inter-node cluster coordination rides the shared store connection and carries **no separate node-to-node socket**, so encrypting the store connection encrypts the cluster traffic with it. Raw TCP and X12 listeners have no transport TLS of their own — keep them on loopback or front them with a TLS-terminating proxy or an isolated trusted segment. This is deployment guidance for the environment, not a limitation of the HA model.

Related

- [CLUSTERING.md](#) — clustering reference.
- [DEPLOYMENT.md](#) — deployment and network-exposure guide.
- [CONFIGURATION.md](#) — the full store and cluster settings catalog.